Database Management - Notes

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Session 1: Data Management

What is a Database

A database is an organized and structured set of records stored and managed for a common purpose. A data record is an information element which, as a unit, describes a complex set of facts.

Architecture and Components of an Information System

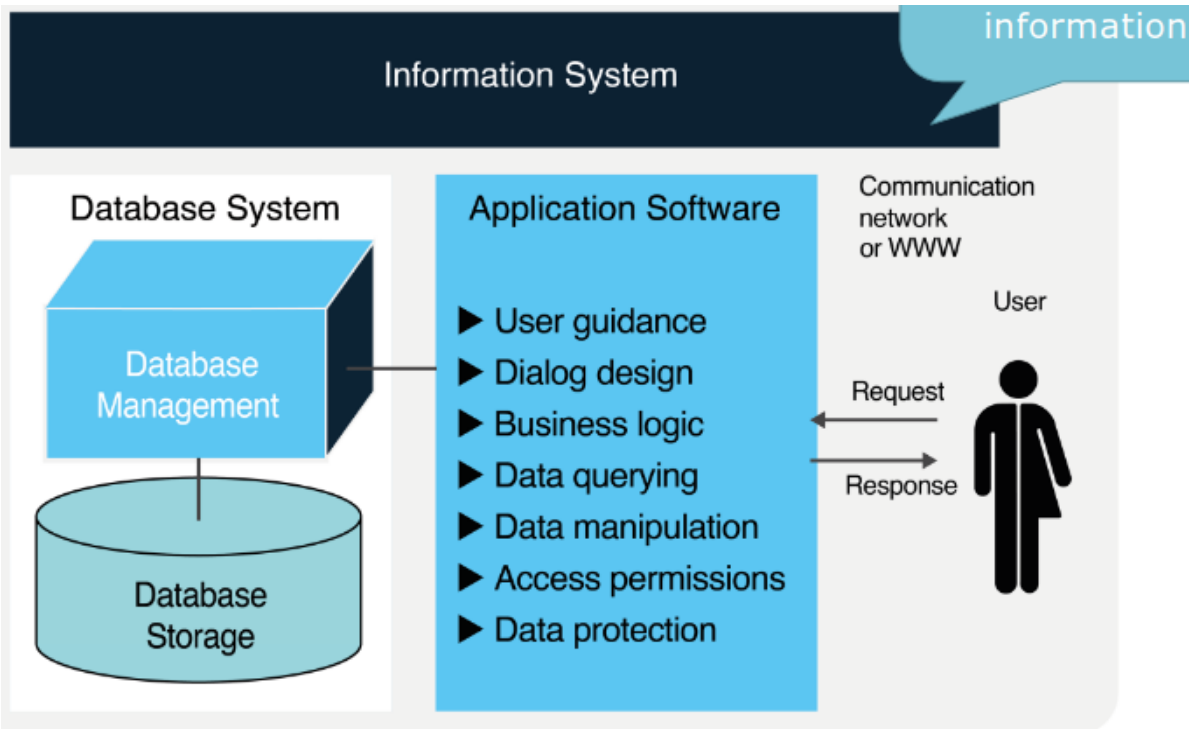

Figure 1: Information System

File System

A file system is a structured method used by an operating system to organize, store, and retrieve data on a storage drive. It manages how data is named, placed, and grouped into files and directories, acting as a bridge between the raw hardware and the user.

Why do we use databases?

- Enable multi-user operations: Manage transactions
- Query and manipulate data sets
- Ensure data consistency and integrity automatically
- Retrieve large data quantities (indexing, parallelization) efficiently

Big Data Challenges

Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing software. Data with many entries (rows) offers greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.

- **Volume**: This refers to the sheer amount of data generated every second, requiring massive storage capacities that go far beyond traditional hardware.
- **Velocity**: This represents the incredible speed at which data is created, processed, and analyzed, often in real-time or near real-time.
- **Variety**: This describes the diverse types of data available, ranging from structured text in databases to unstructured formats like video, images, and social media posts.
- **Veracity**: This focuses on the quality, accuracy, and trustworthiness of the data, as "messy" data can lead to incorrect conclusions.
- **Value**: This is the most critical V, referring to the ability to turn bulk data into meaningful insights that drive business decisions or solve problems.

NoSQL Databases

Optimization for Big Data:

- Non-relational data structures
- Non-relational database languages

Data Management

Data Management comprises all methods related to creating value from data as a resource.

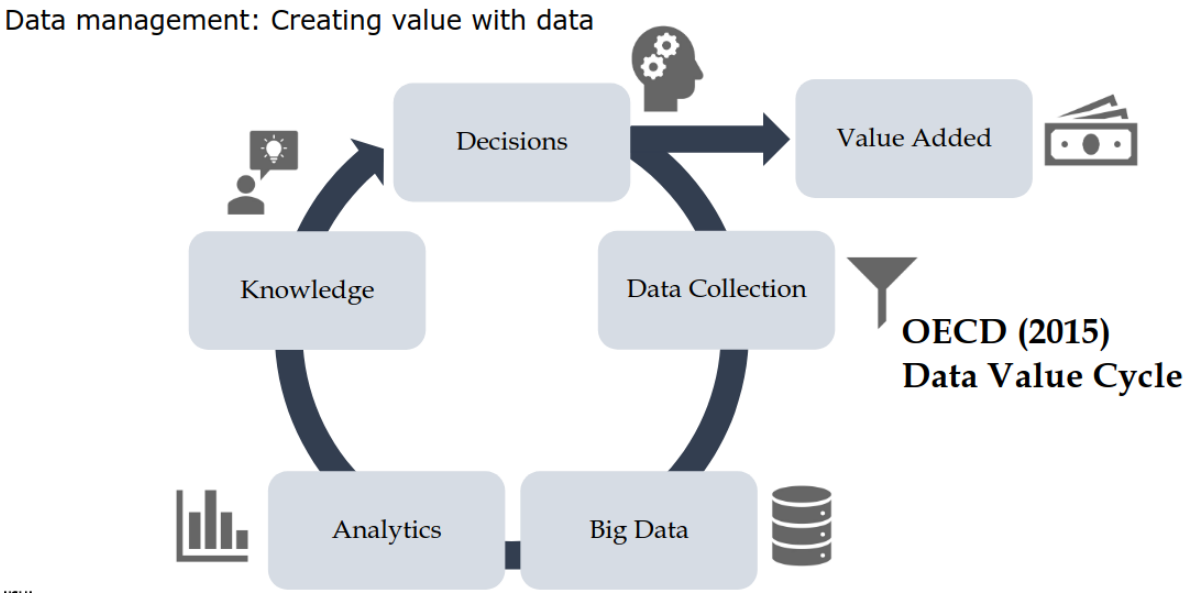

Figure 2: Data Value Cycle

Database Application

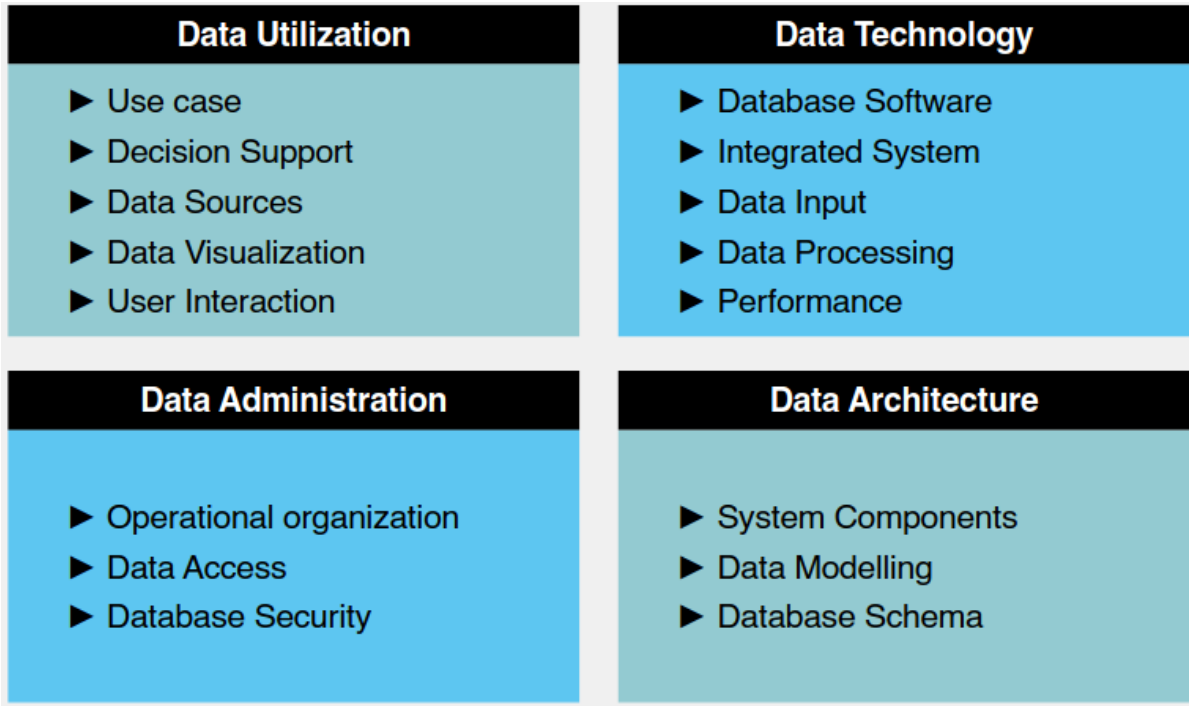

Figure 3: Database Applications